List of formulas

- Equilibrium equations:

$$\sum F_x = 0, \sum F_y = 0, \sum F_z = 0, and\ \sum_{about\ any\ point} Moment = 0$$

- Statically determinacy:

m + r = 2j ➔ Statically determinate
m + r < 2j ➔ Mechanism
m + r > 2j ➔ Statically indeterminate

week3: pin jointed truss

$\sigma = \frac{P}{A}$

- Stress σ = Force/Area, Strain $\varepsilon = \frac{\delta}{L}$ (P above Force, A above Area)
- Modulus of elasticity $E = \frac{\sigma}{\varepsilon}$
- The change in length, δ, due to a change in temperature, ΔT, is given by:` $\delta = \alpha$. Lo . ΔT
- Centre of Area $\bar{y}$

$$\bar{y} = \frac{\int_A y\, dA}{A}$$

- Second moment of area I

$$I = \int_A y^2 dA$$

For rectangle $\bar{y} = 0$ and $I = \frac{bd^3}{12}$

For triangle $\bar{y} = \frac{2d}{3}$ and $I = \frac{bd^3}{4}$

Parallel axis theorem

$I_{uu} = I_{xx} + Ae^2$

- Relationship between bending moment, stress and curvature

$$\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$$

- Deflection of beams

$$EI\frac{d^2v}{dx^2} = -M$$

- Stress due to shear force

$$\tau = F\frac{A\bar{y}}{Ib}$$

Shear flow $q = \frac{FA\bar{y}}{I}$